复旦数学资料汇编

Riemann 几何

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

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第一章　Riemann 几何

1.1　习题: Riemann 几何

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2024 春，丁琪在课堂上布置的习题。

1. Let $F\colon M^n \to \mathbb{R}^N$ be an immersion. F induces a metric $F^*(ds^2)$ on M, as well as a connection

$$\nabla^M_Y X := \sum_{i=1}^n \langle \nabla_Y X, e_i\rangle e_i$$

on M, where $e_1, \cdots, e_n$ is a unitary frame on M. Verify that ∇^M is the Levi-Civita connection on M.

Pf. The problem is a special case of Exercise 3 of Chapter 2, of which we use the notation and provide a proof here. That is mainly to check

(i) let $f \in C^\infty(M)$ be extended arbitrarily to $\overline{M}$, using $\overline{X}f = Xf$, then

$$\nabla^\top_X(fY) = \big(\overline{\nabla}_{\overline{X}}(f\overline{Y})\big)^\top = \big(f\overline{\nabla}_{\overline{X}}\overline{Y}\big)^\top + (\overline{X}f)\overline{Y} = f\nabla^\top_X Y + (Xf)Y.$$

(ii) $\nabla^\top$ is also symmetric, i.e. (the last equality is since $[X,Y]$ is tangent to M as long as X, Y are)

$$\nabla^\top_X Y - \nabla^\top_Y X = (\overline{\nabla}_{\overline{X}}\overline{Y} - \overline{\nabla}_{\overline{Y}}\overline{X})^\top = [\overline{X},\overline{Y}]^\top = [X,Y].$$

(iii) $\nabla^\top$ is compatible with the induced metric, i.e.

$$\begin{aligned} X\langle Y,Z\rangle_M &= \overline{X}\langle \overline{Y},\overline{Z}\rangle_{\overline{M}} = \langle \overline{\nabla}_{\overline{X}}\overline{Y},\overline{Z}\rangle_{\overline{M}} + \langle \overline{Y},\overline{\nabla}_{\overline{X}}\overline{Z}\rangle_{\overline{M}} \\ &= \langle (\overline{\nabla}_{\overline{X}}\overline{Y})^\top, Z\rangle_M + \langle Y, (\overline{\nabla}_{\overline{X}}\overline{Z})^\top\rangle_M \\ &= \langle \nabla^\top_X Y, Z\rangle_M + \langle Y, \nabla^\top_X Z\rangle_M. \end{aligned}$$

2. Show that $R(X,Y)(fZ) = fR(X,Y)Z$.

Pf. We have

$$\begin{aligned} \nabla_Y\nabla_X(fZ) &= \nabla_Y\big(f\nabla_X Z + (Xf)Z\big) \\ &= f\nabla_Y\nabla_X Z + (Yf)\nabla_X Z + (Xf)\nabla_Y Z + (Y(Xf))Z. \end{aligned}$$

Therefore,

$$\nabla_Y\nabla_X(fZ) - \nabla_X\nabla_Y(fZ) = f(\nabla_Y\nabla_X - \nabla_X\nabla_Y)Z + ((YX - XY)f)Z,$$

hence

$$\begin{aligned} R(X,Y)(fZ) &= f(\nabla_Y\nabla_X - \nabla_X\nabla_Y)Z + ([Y,X]f)Z \\ &\quad + ([X,Y]f)Z + f\nabla_{[X,Y]}Z = fR(X,Y)Z. \end{aligned}$$

3. 证明 [Xin] 第 7 章最后的附注 (p.45)。

Pf. 见 [Jo] Theorem 5.5.2 (pp.231–3)。

4. 验证 B^3 在双曲度量下的截面曲率是 -1。

Pf. 见 [Xin] 命题 10.2 (pp.53–4)，最后一步用到了 [dC] Lemma 3.4 (p.96)。

5. 在双曲空间的上半空间模型下，计算它的截面曲率，并确定它的测地线。

Pf. 见 [dC] Chapter 8 section 3 (pp.160–2).

6. 对于 [Xin] 第 11 章定义的 Laplace 算子 (p.65)，验证其局部表达式为（记 $g = \det(g_{ij})$）

$$\triangle f = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}g^{ij}\partial_j f).$$

Pf.（直接计算）由于对 $X = X^i\partial_i, Y = Y^j\partial_j$，有

$$\begin{aligned} \nabla_Y X &= Y^j\nabla_{\partial_j}(X^i\partial_i) = Y^j(\partial_j X^i\partial_i + X^i\nabla_{\partial_j}\partial_i) \\ &= Y^j(\partial_j X^i\partial_i + X^i\Gamma_{ij}^k\partial_k) \\ &= Y^j(\partial_j X^k + X^i\Gamma_{ij}^k)\partial_k, \end{aligned}$$

得到

$$\begin{aligned} \operatorname{div} X &= \operatorname{tr}(Y \mapsto \nabla_Y X) \\ &= \partial_j X^j + X^i\Gamma_{ij}^j \\ &= \partial_j X^j + X^i g^{jk}\partial_i g_{jk}/2, \end{aligned}$$

其中

$$\begin{aligned} \Gamma_{ij}^j &= g^{jk}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij})/2 \\ &= g^{jk}\partial_i g_{jk}/2. \quad (\text{对称的} j,k \text{ 消去}) \end{aligned}$$

用第 9 题，进而有

$$\begin{aligned} \operatorname{div} X &= \partial_j X^j + X^i\partial_i(\ln g)/2 \\ &= \partial_j X^j + X^i\partial_i(\ln\sqrt{g}) \\ &= (1/\sqrt{g})\partial_i(X^i\sqrt{g}). \end{aligned}$$

接着，为了 $Xf = \langle \operatorname{grad} f, X\rangle$，即

$$X^j\partial_j f = g_{ij}(\operatorname{grad} f)^i X^j,$$

得出

$$\operatorname{grad} f = g^{ij}\partial_j f\partial_i.$$

最后，结合两表达式得

$$\triangle f = \operatorname{div}(\operatorname{grad} f) = (1/\sqrt{g})\partial_i(\sqrt{g}g^{ij}\partial_j f).$$

（对偶法）幺正坐标系下，[Xin] 与上式均有 $\triangle f = \sum \partial_{ii} f$，故说明上式不取决于坐标系的选取即可。这只需注意到，对任何局部坐标区域 U，对于测试函数 $\phi \in C_c^\infty(U)$，

$$\int_U \triangle f \phi \sqrt{g} dx = \int_U \partial_i(\sqrt{g}g^{ij}\partial_j f)\phi dx = -\int_U g^{ij}\partial_j f \partial_i \phi \sqrt{g} dx,$$

$g^{ij}\partial_i \otimes \partial_j$ 是 $g_{ij}dx^i \otimes dx^j$ 的对偶，所以在不同坐标系下是相同的；$\sqrt{g}dx$ 亦同坐标系无关。再用 Riesz 表示定理，得 $\triangle f$ 在 U 上无关乎坐标系。

7. 设 M 是完备单连通的 Riemann 流形，其截面曲率 $\leq \kappa$，$\kappa > 0$。设 $o \in M$，距离函数 $\rho(x) = d(o,x)$，那么 $\rho^2(x)$ 在 $B_{\frac{\pi}{\sqrt{\kappa}}}(o)$ 上是光滑凸的。

Pf. 基本是 [Xin] 定理 11.3 的证明，除了改成

$$L''(0) \geq \int_0^r |\dot{U}^\perp|^2 - \kappa |U^\perp|^2 dt.$$

上式对 $r \leq \frac{\pi}{\sqrt{\kappa}}$ 非负，是用 $U^\perp(r) = 0$ 与 Wirtinger 不等式，后者见 [dC] Exercise 11.2 (p.250)。

8. 设 Riemann 流形 M 的截面曲率 K 满足 $0 < L \leq K \leq H$，其中 H 和 L 是常数。设 γ 是 M 中的测地线，那么，γ 上相邻共轭点沿着 γ 的距离 d 满足

$$\frac{\pi}{\sqrt{H}} \leq d \leq \frac{\pi}{\sqrt{L}}.$$

Pf. 见 [dC] Chapter 10 Proposition 2.4 (p.218)。

9. 设方阵 $A = (a_{ij})_{1\leq i,j\leq n}$，其元素 a_{ij} 均是连续可微的 t 的函数。验证 $\partial_t\big(\ln\det A\big) = \sum\limits_{i,j=1}^{n} a^{ij}\partial_t a_{ij}$。

Pf. 记 $n-1$ 阶方阵 $A_{ij} = (-1)^{k+l}(a_{kl})_{k\neq i, l\neq j}$ 是 a_{ij} 的代数余子式，则

$$\partial_t(\det A) = \sum_{i,j=1}^{n} \det A_{ij}\partial_t a_{ij} = \sum_{i,j=1}^{n} \det A\, a^{ij}\partial_t a_{ij}.$$

第一个等号是 Leibniz 法则，第二个是由于 $AA^* = (\det A)I$ 说明 $A_{ij} = \det A\, a^{ij}$，其中伴随矩阵 $A^* = (A_{ij})_{1\leq i,j\leq n}$。

10. 所谓悬链面（catenoid），是 $\mathbb{R}^3$ 中的曲面

$$\Sigma = \{x_1^2 + x_2^2 = \cosh^2 x_3\}.$$

(i) 证明 Σ 是极小的。(ii) 计算 Gauss 曲率的积分 $\int_\Sigma K\, dV$。(iii) 证明 Σ 上的正的上调和函数是常数。

注：$\cosh x = (e^x + e^{-x})/2$，$\sinh x = (e^x - e^{-x})/2$.

Pf. (i) Σ 是 $r(u,v) = (\cosh v \cos u, \cosh v \sin u, v)$ 的图像，有

$$r_u = (-\cosh v \sin u, \cosh v \cos u, 0),$$
$$r_v = (\sinh v \cos u, \sinh v \sin u, 1),$$

从而 $dr^2 = Edu^2 + 2Fdudv + Gdv^2$ 的分量为

$$\begin{aligned} E &= |r_u|^2 = \cosh^2 v, \\ F &= r_u \cdot r_v = 0, \\ G &= |r_v|^2 = \sinh^2 v + 1 = \cosh^2 v, \end{aligned}$$

故 r 是等温坐标。由 [dC'] p.204 的 corollary，其三个坐标分量都是调和函数表明 Σ 是极小的。

(ii) 对于 2 维流形上的共形度量 $g = \rho^2 \tilde{g}$，成立 Yamabe 方程

$$K = [\tilde{K} - \triangle(\ln \rho)]/\rho^2,$$

其中 $\tilde{K}$ 为 $\tilde{g}$ 的 Gauss 曲率（见 [YS] 第五章）。由此，

$$K = -[\ln(\cosh v)]''/\cosh^2 v = -\cosh^{-4} v.$$

故

$$\begin{aligned} \int_\Sigma K\, dV &= -2\pi \int_{\mathbb{R}} \frac{\cosh^2 v\, dv}{\cosh^4 v} \\ &= -2\pi \int_{\mathbb{R}} \frac{dv}{[(e^v + e^{-v})/2]^2} \\ &= -4\pi \int_0^\infty \frac{d(e^{2v})}{(e^{2v}+1)^2} = -4\pi. \end{aligned}$$

(iii) $r\colon \mathbb{R}^2 \to \Sigma$ 诱导了 Σ 上的 Riemann 度量 $dr^2 = \cosh^2 v(du^2 + dv^2)$，由此

$$\triangle_\Sigma = \cosh^{-2} v \triangle_{\mathbb{R}^2}.$$

故设 $f\colon \Sigma \to \mathbb{R}$ 是上调和函数，则 $f \circ r\colon \mathbb{R}^2 \to \mathbb{R}$ 同样是上调和函数。由 Hadamard 三圆定理，可以说明 $\mathbb{R}^2$ 上的正的上调和函数是常数。

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1.2 习题: do Carmo

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Differentiable Manifolds

2. Prove that the tangent bundle of a differentiable manifold is orientable (even though M may not be).
   Let π be the projection from TM to M. Cover M with $\{(U,\varphi)\}$, then $\{\pi^{-1}(U)\}$ covers $TM = \pi^{-1}(M)$, and we need to find the trivialization $\tilde{\varphi}$ associated to $\pi^{-1}(U)$ for each U.
   Given any smooth chart $\varphi\colon U \subset M \to \mathbb{R}^n$, define $\tilde{\varphi}\colon \pi^{-1}(U) \subset TM \to \mathbb{R}^{2n}$ by

   $$v|_p \mapsto (\varphi(p), v).$$

   For any two overlapping charts (U,φ) and (V,ψ), the transition map $\tilde{\psi}\circ\tilde{\varphi}^{-1}\colon \varphi(U\cap V)\times\mathbb{R}^n \to \psi(U\cap V)\times\mathbb{R}^n$ is given by

   $$(x, v) \mapsto (\tilde{x}, \frac{\partial \tilde{x}}{\partial x} v),$$

   where $\psi\circ\varphi^{-1}(x) := \tilde{x}(x) = (\tilde{x}^1(x), \cdots, \tilde{x}^n(x))$. Then the determinant is $\left(\det \frac{\partial \tilde{x}}{\partial x}\right)^2 > 0$.

Riemannian Metrics

2. In terms of $\tilde{\pi}\colon \mathbb{R}^n \xrightarrow{\pi} T^n \hookrightarrow \mathbb{R}^{2n}$, which is,

   $$\tilde{\pi}(x_1, \cdots, x_n) = (\cos x_1, \sin x_1, \cdots, \cos x_n, \sin x_n),$$

   we have $d\tilde{\pi} = \mathrm{diag}((-\sin x_1, \cos x_1), \cdots, (-\sin x_n, \cos x_n))$. So the induced metric on $\mathbb{R}^n$ is

   $$(-\sin x_j du^j)\otimes(-\sin x_j du^j) + (\cos x_j du^j)\otimes(\cos x_j du^j) = du^j \otimes du^j.$$

4. (a) For $g(t) = yt + x$ and $f(t) = bt + a$,

$$(g \circ f)(t) = y(bt + a) + x = ybt + ya + x.$$

That is to say, L_g carries (a, b) to $(ya + x, yb)$, and thus

$$dL_g = \begin{pmatrix} y & 0 \\ 0 & y \end{pmatrix}.$$

As $L_g(0, 1) = (x, y)$, we have

$$\langle -, - \rangle_{(0,1)} = \langle (dL_g)-, (dL_g)- \rangle_{(x,y)} = y^2 \langle -, - \rangle_{(x,y)}.$$

(b) Check that $-\dfrac{4dzd\bar{z}}{(z - \bar{z})^2}$ is invariant under $z \mapsto az + b$ and $z \mapsto 1/z$.

6. Exercise 2 provides a local isometry $\pi \colon \mathbb{R}^n \to T^n$, but there is no local isometry $\rho \colon T^n \to \mathbb{R}^n$. Indeed, write $\rho = (\rho_1, \cdots, \rho_n)$. ρ_1 mapping the compact set T^n to $\mathbb{R}$ attains its maximum at some $p \in T^n$, and consequently $d\rho_1(p) = 0 \in \mathbb{R}^n$. Then $d\rho(p) = (0, \cdots) \in \mathbb{R}^{n \times n}$ is singular.

Affine and Riemannian Connections

1. Denote P by P_t instead. Since

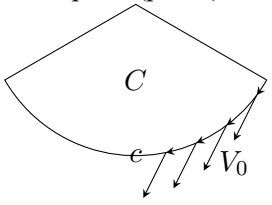
$$\frac{d}{dt}\langle P_t v, P_t w \rangle = \left\langle \frac{D}{dt}(P_t v), P_t w \right\rangle + \left\langle P_t v, \frac{D}{dt}(P_t w) \right\rangle = 0,$$

P_t is an isometry. Then $\det P_t \in \{1, -1\}$. Because P_t is continuously dependent on t and $P_{t_0} = \mathrm{id}$, we have $\det P_t$ is always 1, and in particular positive.

2. Proved in [CL] Chapter 2 Theorem 7.2 (pp.142-3).

4. a) According to Exercise 3, $\nabla_{\dot{c}} V$ is the tangential component of $\overline{\nabla}_{\dot{c}} V$, i.e. the orthonormal projection from $T_{c(t)}\mathbb{R}^3$ onto $T_{c(t)}M$, where $\overline{\nabla}$ is the Riemannian connection of $\mathbb{R}^3$. $\overline{\nabla}_{\dot{c}} V$ is simply the usual derivative $\frac{dV}{dt}$, so $\nabla_{\dot{c}} V = 0$ iff the projection of $\frac{dV}{dt}$ onto $T_{c(t)}M$ vanishes.
b) The acceleration of the unit velocity field along a great circle, namely $\frac{dV}{dt}$, points to the center of the sphere, perpendicular to the tangent space embedded in $\mathbb{R}^3$ (or $\mathbb{R}^{n+1}$).

7. Method I. Consider the cone C tangent to S^2 along c, then the parallel transport of V_0 along c is the same, whether taken relative to S^2 or to C, as the tangent spaces along c whether taken relative to S^2 or to C are the same. Flatten the cone (locally, or at least minus a line) to a sector on the plane (in other words, C is locally isometric to a sector in $\mathbb{R}^2$ with c being its arc). The parallel transport of V_0 on the plane is merely translation in $\mathbb{R}^2$. From the perspective of a person walking along c, V_0 appears to be rotating at constant speed, reminiscent of a scene when the theme song of *Detective Conan* goes '運命のルーレット廻して、ずっと君を見ていた'. For details, refer to [dC] 4-4 Example 1 (p.246).



Method II. Consider the cone C tangent to S^2 along c, parametrized as

$$\{(av\cos u, av\sin u, bv) \in \mathbb{R}^3 : u, v \in \mathbb{R}\},$$

where $a^2 + b^2 = 1$. The parallel transport of V_0 along c is the same, whether taken relative to S^2 or to C, as the tangent spaces along c whether taken relative to S^2 or to C are the same. Now we carry out a bit of calculations to see the effect of parallel–translating V_0 along c, written $U(t)\partial_u + V(t)\partial_v$. Besides, let $u(t) = t, v(t) = v_0$ be the local expression for $c(t)$. First,

$$g_{uu} = a^2v^2, \quad g_{uv} = 0, \quad g_{vv} = 1,$$

and then

$$\Gamma^u_{uu} = 0, \quad \Gamma^u_{vu} = \frac{1}{v}, \quad \Gamma^v_{uu} = -a^2v, \quad \Gamma^v_{vu} = 0.$$

Substituting into

$$\begin{cases} U + (\Gamma^u_{uu}\dot{u} + \Gamma^u_{uv}\dot{v})U + (\Gamma^u_{vu}\dot{u} + \Gamma^u_{vv}\dot{v})V = 0, \\ V + (\Gamma^v_{uu}\dot{u} + \Gamma^v_{uv}\dot{v})U + (\Gamma^v_{vu}\dot{u} + \Gamma^v_{vv}\dot{v})V = 0, \end{cases}$$

where $\dot{u} = 1, \dot{v} = 0$, we get

$$\begin{cases} U + \dfrac{1}{v}V = 0, \\ V - a^2vU = 0. \end{cases}$$

Hence,

$$\begin{cases} U = \ c\ \cos(at + d), \\ V = av_0\sin(at + d) \end{cases}$$

for two constants c, d. We conclude that the vector rotates at angular velocity a, if the angular velocity of c along the parallel is taken to be 1.

Geodesics; Convex Neighborhoods

1. a) The induced metric is

$$\big(d(f(v)\cos u)\big)^2 + \big(d(f(v)\sin u)\big)^2 + \big(d(g(v))\big)^2 = f^2du^2 + (f'^2 + g'^2)dv^2.$$

b) The inverse is

$$g^{11} = \frac{1}{f^2}, \quad g^{12} = 0, \quad g^{22} = \frac{1}{f'^2 + g'^2}.$$

Then the coefficients in these differential equations are

$$\begin{aligned} \Gamma^1_{12} + \Gamma^1_{21} = 2\Gamma^1_{12} &= g^{k1}(\partial_1 g_{2k} + \partial_2 g_{1k} - \partial_k g_{12}) = g^{11}\partial_2 g_{11} = \frac{2ff'}{f^2}, \\ \Gamma^2_{11} &= \frac{1}{2}g^{k2}(\partial_1 g_{1k} + \partial_1 g_{1k} - \partial_k g_{11}) = -\frac{1}{2}g^{22}\partial_2 g_{11} = -\frac{ff'}{f'^2 + g'^2}, \\ \Gamma^2_{22} &= \frac{1}{2}g^{k2}(\partial_2 g_{2k} + \partial_2 g_{2k} - \partial_k g_{22}) = \frac{1}{2}g^{22}\partial_2 g_{22} = \frac{f'f'' + g'g''}{f'^2 + g'^2}. \end{aligned}$$

c) Let

$$\gamma(t) = \big(f(v(t))\cos u(t), f(v(t))\sin u(t), g(v(t))\big),$$

then

$$|\gamma'(t)|^2 = f^2u'^2 + (f'^2 + g'^2)v'^2.$$

Thus by the first (which gives $f^2u'' = -2ff'u'v'$) and the second equation subsequently,

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}|\gamma'(t)|^2 &= f^2u'u'' + ff'v'u'^2 + (f'^2 + g'^2)v'v'' + (f'f'' + g'g'')v'^3\\ &= -2ff'u'^2v' + ff'v'u'^2 + (f'^2 + g'^2)v'v'' + (f'f'' + g'g'')v'^3\\ &= -ff'v'u'^2 + (f'^2 + g'^2)v'v'' + (f'f'' + g'g'')v'^3\\ &= v'(f'^2 + g'^2)\bigg(-\frac{ff'}{f'^2 + g'^2}u'^2 + v'' + \frac{f'f'' + g'g''}{f'^2 + g'^2}v'^2\bigg) = 0.\end{aligned}$$

We have

$$r = \text{radius of } \{\big(f(v)\cos u, f(v)\sin u, g(v)\big) : u\} = f(v),$$

and $\gamma' = u'\partial_1 + v'\partial_2$,

$$r\cos\beta = r\frac{\langle\gamma', \partial_1\rangle}{|\gamma'||\partial_1|} = f\frac{u'f^2}{|\gamma'|f} = \frac{u'f^2}{|\gamma'|}.$$

That $|\gamma'|$ is constant has been verified, and

$$(u'f^2)' = u''f^2 + u'2ff'v' = 0$$

shows that $u'f^2$ is constant as well.

d) Click, or cf. [dC] 4-4 Example 6 (pp.261-2), or [CL] Chapter 3 Exercise 2 (pp.465-6).

2. a) The inner product being well–defined is because

$$d\pi(V) = p'(0), \quad \frac{Dv}{dt}(0) = v'(0) + p_i'(0)v^{j\prime}(0)\nabla_{\partial_i}\partial_j$$

depends only on $V = (p'(0), v'(0)) = (p_i'(0), v^{j\prime}(0))$, not essentially on α. To show that it is a Riemannian metric, properties like symmetry, bilinearity, non-negativity and smooth dependence are all obvious, and $\langle V, V\rangle = 0$ implies $p'(0) = \dfrac{Dv}{dt}(0) = 0$, which further implies $v'(0) = 0$ by the expression of $\dfrac{Dv}{dt}(0)$ in local coordinates above.

b) We have the decomposition

$$T_{(p,v)}TM \simeq T_v(T_pM) \oplus \pi^{-1}(p) \simeq T_pM \oplus T_pM,$$

and $(p'(t), v'(t)) \perp \pi^{-1}(p)$ iff $\dfrac{Dv}{dt} = 0$, i.e. v is parallel along p.

c) Simply because γ' is parallel along γ.

d) The hint is a solution indeed.

e) With respect to the decomposition given in b), $W = d\pi(W) + 0$ if W is horizontal, and $W = 0 + W = 0 + \dfrac{Dw}{dt}(0)$ if W is vertical.

4. a) Write A for the convex neighborhood and a for any of its points. By definition, for any $b \in A$, there is a unique geodesic $\gamma_b\colon [0,1] \to A$ joining a to b. Then $h\colon [0,1]\times A \to A, (t,b) \mapsto \exp_a(t\gamma_b')$ is a homotopy between k_a and id_A.

b) For any $p \in M$ pick a convex neighborhood. A finite union of convex neighborhoods is again convex (check by definition), hence contractible.

7. Let U be a normal neighborhood of p in M, that is, $\exp_p\colon V \to U$ is a diffeomorphism. Define $E_i(\exp_p(v)) = (d\exp_p)_v(\partial_i)$, where $v \in V \subset T_pM$, $\partial_i = \frac{\partial}{\partial x_i}$, and $x_i, i = 1, \cdots, n$ are the standard Euclidean coordinates on $T_pM \simeq \mathbb{R}^n$. In other words, $(d\exp_p)_v$ maps ∂_i at v to E_i at $\exp_p(v)$.

   For any $v = (v_i)_{i=1,\cdots,n} \in V$, $\gamma(t) = \exp_p(tv), t \in [0,1]$ is a geodesic. Then $(\nabla_{\gamma'}\gamma')(\exp_p(tv)) = 0$, where $\gamma' = \big(\exp_p(tv)\big)' = \sum\limits_i v_i E_i(\exp_p(tv))$. In particular let $t = 0$, we get $(\nabla_{\gamma'}\gamma')(p) = 0$ where $\gamma' = \sum\limits_i v_i E_i(p)$, and consequently $\nabla_{E_i}E_j(p) = 0$ by the arbitrariness of v.

8. a) The first is because

$$\langle \operatorname{grad} f, \sum_i v_i E_i\rangle = df(\sum_i v_i E_i) = \sum_i v_i E_i(f) = \langle \sum_i E_i(f)E_i, \sum_i v_i E_i\rangle.$$

   For the second, for $Y = \sum\limits_j g_j E_j$, the trace of the mapping

$$\sum_j g_j E_j \mapsto \nabla_{\sum\limits_j g_j E_j} \sum_i f_i E_i = \sum_{i,j} g_j E_j(f_i) E_i$$

   is $\operatorname{div} X = \sum\limits_i E_i(f_i)$.

   b) Apply a), as $\frac{\partial}{\partial x_1}, \cdots, \frac{\partial}{\partial x_n}$ is a geodesic frame.

10. Differentiate $\left\langle \frac{\partial f}{\partial s}, \frac{\partial f}{\partial t} \right\rangle$ with respect to s, obtaining

$$\begin{aligned}\frac{d}{ds}\left\langle \frac{\partial f}{\partial s}, \frac{\partial f}{\partial t} \right\rangle &= \left\langle \frac{D}{ds}\frac{\partial f}{\partial s}, \frac{\partial f}{\partial t} \right\rangle + \left\langle \frac{\partial f}{\partial s}, \frac{D}{ds}\frac{\partial f}{\partial t} \right\rangle \\ &= 0 + \left\langle \frac{\partial f}{\partial s}, \frac{D}{dt}\frac{\partial f}{\partial s} \right\rangle \\ &= \frac{1}{2}\frac{d}{dt}\left\langle \frac{\partial f}{\partial s}, \frac{\partial f}{\partial s} \right\rangle = 0,\end{aligned}$$

   we get

$$\left\langle \frac{\partial f}{\partial s}, \frac{\partial f}{\partial t} \right\rangle (s_0, t_0) = \left\langle \frac{\partial f}{\partial s}, \frac{\partial f}{\partial t} \right\rangle (0, t_0) = 0.$$

Curvature

1. a) Combine

$$\nabla_X Y + \nabla_Y X = \nabla_{(X+Y)}(X+Y) - \nabla_X X - \nabla_Y Y = 0$$

   and

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

   b) Using a) and the Jacobi identity,

$$\begin{aligned}R(X,Y)Z &= \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z + \nabla_{[X,Y]}Z \\ &= \frac{1}{2}\big(\nabla_Y[X,Z] - \nabla_X[Y,Z] + [[X,Y],Z]\big) \\ &= \frac{1}{4}\big([Y,[X,Z]] - [X,[Y,Z]]\big) + \frac{1}{2}[[X,Y],Z] \\ &= \frac{1}{4}[[X,Y],Z].\end{aligned}$$

c) By the equation (3) of Chapter 1 section 2 (p.40), we have

$$K(\sigma) = \langle R(X,Y)X, Y\rangle = \frac{1}{4}\langle [[X,Y],X],Y\rangle$$
$$= -\frac{1}{4}\langle [X,Y],[Y,X]\rangle = \frac{1}{4}\|[X,Y]\|^2.$$

2. a) $0 = Z\langle X, X\rangle = 2\langle A_X(Z), X\rangle$。

b) Let S be the right side, that is to say,

$$S = Z\langle \nabla_Z X, X\rangle + \langle \nabla_Z \nabla_X X, Z\rangle - \langle \nabla_X \nabla_Z X, Z\rangle + \langle \nabla_{[X,Z]} X, Z\rangle.$$

Using the Killing equation $\langle \nabla_Z X, X\rangle + \langle \nabla_X X, Z\rangle = 0$, the sum of first two terms becomes

$$Z\langle \nabla_Z X, X\rangle + \langle \nabla_Z \nabla_X X, Z\rangle = -Z\langle \nabla_X X, Z\rangle + \langle \nabla_Z \nabla_X X, Z\rangle = -\langle \nabla_X X, \nabla_Z Z\rangle;$$

note that $\langle \nabla_Z X, Z\rangle = 0$, implying $X\langle \nabla_Z X, Z\rangle = 0$, and then the third term becomes

$$-\langle \nabla_X \nabla_Z X, Z\rangle = \langle \nabla_Z X, \nabla_X Z\rangle;$$

the fourth term is

$$\langle \nabla_{[X,Z]} X, Z\rangle = -\langle \nabla_Z X, [X,Z]\rangle.$$

Adding together, we get

$$S = -\langle \nabla_X X, \nabla_Z Z\rangle + \langle \nabla_Z X, \nabla_Z X\rangle.$$

Because of the Killing equation at p and a),

$$0 = (\langle \nabla_X X, Y\rangle + \langle \nabla_Y X, X\rangle)(p) = \langle \nabla_X X, Y\rangle(p)$$

for any Y, it is inferred that $\nabla_X X(p) = 0$, and therefore

$$S(p) = \langle \nabla_Z X, \nabla_Z X\rangle(p).$$

3. That A is anti–symmetric is simply

$$\langle A(y), z\rangle + \langle y, A(z)\rangle = \langle \nabla_Y X, Z\rangle + \langle Y, \nabla_Z X\rangle = 0,$$

and that A is an isomorphism is because by 2.b),

$$\langle A(y), z\rangle + \langle y, A(z)\rangle(p) = \frac{1}{2} Z_p(Zf) + (X,Z,X,Z)(p) > 0.$$

As $-A = A^{\mathrm{T}}$, taking determinant on both sides gives

$$(-1)^{\dim E} \det A = \det A.$$

This implies $\dim E = \dim M - 1$ is even, which is a contradiction to $\dim M$ being even.

5. Observe that, for all $Z \in \mathscr{X}(M)$, at $\gamma(0)$ we have (where $Z' = \frac{DZ}{dt}$)

$$0 = (\nabla_{\gamma'} R)(\gamma', X, \gamma', Z) = \nabla R(\gamma', X, \gamma', Z, \gamma')$$
$$= \gamma'(R(\gamma', X, \gamma', Z)) - R(\gamma', X', \gamma', Z) - R(\gamma', X, \gamma', Z')$$
$$= \gamma'\langle R(\gamma', X)\gamma', Z\rangle - \langle R(\gamma', X')\gamma', Z\rangle - \langle R(\gamma', X)\gamma', Z'\rangle$$
$$= \langle \nabla_{\gamma'}(R(\gamma', X)\gamma'), Z\rangle - \langle R(\gamma', X')\gamma', Z\rangle.$$

The first equality is because the tensor $\nabla_{\gamma'} R$ is tensorial in the second component and X vanishes at $\gamma(0)$, and the third equality is because $\gamma'' = 0$.

6. a) $\nabla R = 0$ implies that for any W,

$$\begin{aligned}\gamma'(R(X,Y,Z,W)) = R(\nabla_{\gamma'}X,Y,Z,W) + R(X,\nabla_{\gamma'}Y,Z,W)\\ + R(X,Y,\nabla_{\gamma'}Z,W) + R(X,Y,Z,\nabla_{\gamma'}W).\end{aligned}$$

Substituting $\nabla_{\gamma'}X = \nabla_{\gamma'}Y = \nabla_{\gamma'}Z = 0$, we get

$$\gamma'(R(X,Y,Z,W)) = R(X,Y,Z,\nabla_{\gamma'}W),$$

which is $\langle \nabla_{\gamma'}(R(X,Y)Z), W\rangle = 0$.

b) For any curve γ, let X, Y be two orthonormal parallel unit vector fields along γ. Then we have

$$\frac{d}{dt}\big(R(X,Y,X,Y)(\gamma(t))\big) = \langle \nabla_{\gamma'}(R(X,Y)X), Y\rangle + \langle R(X,Y)X, \nabla_{\gamma'}Y\rangle = 0.$$

To put it another way, $R(X,Y,X,Y)$ is constant along γ. Therefore $R(X,Y,X,Y)$ is constant over M.

c) Recall that M has constant sectional curvature K_0 means

$$R(X,Y,Z,W) = K_0(\langle X,W\rangle\langle Y,Z\rangle - \langle Y,W\rangle\langle X,Z\rangle).$$

Then $\nabla R(X,Y,Z,W,V) = 0$, i.e.

$$\begin{aligned}V(R(X,Y,Z,W)) = R(\nabla_V X,Y,Z,W) + R(X,\nabla_V Y,Z,W)\\ + R(X,Y,\nabla_V Z,W) + R(X,Y,Z,\nabla_V W),\end{aligned}$$

becomes the identity

$$\begin{aligned}&V(\langle X,W\rangle\langle Y,Z\rangle - \langle Y,W\rangle\langle X,Z\rangle)\\ &= \langle \nabla_V X,W\rangle\langle Y,Z\rangle - \langle Y,W\rangle\langle \nabla_V X,Z\rangle\\ &+ \langle X,W\rangle\langle \nabla_V Y,Z\rangle - \langle \nabla_V Y,W\rangle\langle X,Z\rangle\\ &+ \langle X,W\rangle\langle Y,\nabla_V Z\rangle - \langle Y,W\rangle\langle X,\nabla_V Z\rangle\\ &+ \langle X,\nabla_V W\rangle\langle Y,Z\rangle - \langle Y,\nabla_V W\rangle\langle X,Z\rangle.\end{aligned}$$

7. In this frame $\nabla_{e_i}e_j(p) = 0$, hence

$$\begin{aligned}\nabla R(e_i,e_j,e_k,e_\ell,e_h) &= e_h\langle R(e_i,e_j)e_k,e_\ell\rangle = e_h\langle R(e_k,e_\ell)e_i,e_j\rangle\\ &= \langle \nabla_{e_h}\nabla_{e_\ell}\nabla_{e_k}e_i - \nabla_{e_h}\nabla_{e_k}\nabla_{e_\ell}e_i, e_j\rangle.\end{aligned}$$

Note that though there is supposed to be one more term $\langle \nabla_{e_h}\nabla_{[e_k,e_\ell]}e_i, e_j\rangle$ in the second line above, it actually vanishes, as long as we observe that

$$[e_k,e_\ell](p) = (\nabla_{e_k}e_\ell - \nabla_{e_\ell}e_k)(p) = 0.$$

In the second Bianchi Identity, (i,j) stays invariant, while $(k,\ell,h)\to(\ell,h,k)\to(h,k,\ell)$ permutes cyclically. Summing these three equations yields

$$\nabla R(e_i,e_j,e_k,e_\ell,e_h) + \nabla R(e_i,e_j,e_\ell,e_h,e_k) + \nabla R(e_i,e_j,e_h,e_k,e_\ell)$$

$$
\begin{aligned}
&= \langle \nabla_{e_h}\nabla_{e_\ell}\nabla_{e_k}e_i - \nabla_{e_h}\nabla_{e_k}\nabla_{e_\ell}e_i, e_j\rangle \\
&+ \langle \nabla_{e_k}\nabla_{e_h}\nabla_{e_\ell}e_i - \nabla_{e_k}\nabla_{e_\ell}\nabla_{e_h}e_i, e_j\rangle \\
&+ \langle \nabla_{e_\ell}\nabla_{e_k}\nabla_{e_h}e_i - \nabla_{e_\ell}\nabla_{e_h}\nabla_{e_k}e_i, e_j\rangle \\
&= R(e_\ell, e_h, \nabla_{e_k}e_i, e_j) + R(e_h, e_k, \nabla_{e_\ell}e_i, e_j) + R(e_k, e_\ell, \nabla_{e_h}e_i, e_j) \\
&= 0,
\end{aligned}
$$

where the last equality is simply since $\nabla_{e_k}e_i = \nabla_{e_\ell}e_i = \nabla_{e_h}e_i = 0$ at p.

Jacobi Fields

3. Assume that $a \in C(p)$, then the existence of a non-trivial Jacobi field J along the geodesic $\gamma\colon [0,a] \to M$, with $\gamma(0) = p, J(0) = J(a) = 0$. Using the Jacobi equation, we have

$$
\frac{d}{dt}\left\langle \frac{DJ}{dt}, J \right\rangle = -(\gamma', J, \gamma', J) + \left\| \frac{DJ}{dt} \right\|^2 \geq 0.
$$

But $\left\langle \frac{DJ}{dt}, J\right\rangle = 0$ at $t = 0, a$, then $\left\langle \frac{DJ}{dt}, J\right\rangle \equiv 0$. Since $\frac{d}{dt}\langle J, J\rangle = 2\left\langle \frac{DJ}{dt}, J\right\rangle \equiv 0$, we get $\|J\|^2 \equiv$ constant, and hence $\equiv 0$, a contradiction.

4. From Example 2.3, the Jacobi field J_1 along γ satisfying $J_1(0) = 0, J_1'(0) = \frac{u_0}{|u_0|}$, is given by

$$
J_1(t) = \frac{\sin(t\sqrt{-b})}{\sqrt{-b}} w(t).
$$

In addition, from Corollary 2.5,

$$
J_1(\ell) = (d\exp_p)_{\ell\gamma'(0)}(\ell w(0)).
$$

It follows that

$$
J(\ell) = v = (d\exp_p)_{\ell\gamma'(0)}(u_0) = J_1(\ell)\frac{|u_0|}{|\ell w(0)|} = J_1(\ell)\frac{|u_0|}{\ell}.
$$

In addition, since $|w(t)| = |w(0)| = 1$,

$$
1 = |v| = |J(\ell)| = |J_1(\ell)|\frac{|u_0|}{\ell} = \frac{\sin(\ell\sqrt{-b})}{\sqrt{-b}}\frac{|u_0|}{\ell},
$$

we have

$$
\frac{|u_0|}{\ell} = \left(\frac{\sin(\ell\sqrt{-b})}{\sqrt{-b}}\right)^{-1},
$$

which implies what was asserted.

6. a) By Proposition 2.9, $\exp_p\colon B_\delta(0) \subset T_pM \to B_\delta(p) \subset M$ is a diffeomorphism onto an open subset $\bar{U}$ of M. To make the polar coordinates (ρ, θ) into a coordinate system indeed, an entire radius $-\rho v(0), 0 \leq \rho < \delta$ in $B_\delta(0)$ must be removed, and correspondingly the ray $\exp_p(-\rho v(0)), 0 \leq \rho < \delta$ in $\bar{U}$, forming the required U.

b) $f(0,\theta), -\pi < \theta < \pi$ is the point p instead of a curve, so $g_{12}(0,\theta) = 0$. Considering the unit–speed geodesic $f(\rho, \theta_0), 0 < \rho < \delta$ where ρ varies and θ_0 is fixed, we get $g_{11} = 1$, and $\nabla_{\partial_\rho}\partial_\rho = 0$ for $\partial_\rho := \frac{\partial f}{\partial \rho}(\rho, \theta_0)$, the latter implying

$$
0 = 2\langle \partial_\theta, \nabla_{\partial_\rho}\partial_\rho\rangle = 2\partial_\rho\langle \partial_\rho, \partial_\rho\rangle - \partial_\theta\langle \partial_\rho, \partial_\rho\rangle,
$$

i.e.

$$0 = 2\partial_1 g_{12} - \partial_2 g_{11}.$$

So $\partial_1 g_{12} = 0$, and consequently $g_{12} = 0$. $g_{22} = \left|\frac{\partial f}{\partial \theta}\right|^2$ is too simple to be explained.

c) On one hand, according to b) and Corollary 2.10,

$$\sqrt{g_{22}} = \left|\frac{\partial f}{\partial \theta}\right| = |J(\rho)| = \rho - \frac{1}{6}K(p)\rho^3 + o(\rho^3);$$

on the other hand, the Taylor expansion yields

$$\sqrt{g_{22}} = \left[\sqrt{g_{22}} + (\sqrt{g_{22}})_\rho \rho + \frac{1}{2}(\sqrt{g_{22}})_{\rho\rho}\rho^2 + \frac{1}{6}(\sqrt{g_{22}})_{\rho\rho\rho}\rho^3\right]\Big|_{(0,0)} + o(\rho^3).$$

Comparing the coefficients, we get

$$\sqrt{g_{22}} = 0, \quad (\sqrt{g_{22}})_\rho = 1, \quad (\sqrt{g_{22}})_{\rho\rho} = 0, \quad (\sqrt{g_{22}})_{\rho\rho\rho} = -K(p)$$

at $(0,0)$, and then

$$(\sqrt{g_{22}})_{\rho\rho} = \left[(\sqrt{g_{22}})_{\rho\rho} + (\sqrt{g_{22}})_{\rho\rho\rho}\rho\right]\Big|_{(0,0)} + o(\rho) = -K(p)\rho + o(\rho).$$

d)

$$\lim_{\rho\to 0}\frac{(\sqrt{g_{22}})_{\rho\rho}}{\sqrt{g_{22}}} = \lim_{\rho\to 0}\frac{-K(p)\rho + o(\rho)}{\rho + o(\rho)} = -K(p).$$

7. Using Exercise 6 c), we have

$$L_r = \int_0^{2\pi}\left|\frac{\partial f}{\partial \theta}\right| d\theta = \int_0^{2\pi}(-K(p)\rho + o(\rho^3))\, d\theta = 2\pi r - \frac{\pi}{3}K(p)r^3 + o(r^3).$$

Isometric Immersions

1. a) Trivially verified by definition or local expressions.

   b) Suppose that $\gamma_p = \{p\} \times \gamma$ is a geodesic of $(M_2)_p$, naturally identified with γ which is a geodesic of M_2. That is to say, $\nabla^2_{\gamma'}\gamma' = 0$. Then using a), we too have $\nabla_{\gamma_p'}\gamma_p' = \nabla^1_0 0 + \nabla^2_{\gamma'}\gamma' = 0$.

   c) Easy to see that x, y commute, and so do ∇^1_x, ∇^2_y. Hence $R(x,y) = [\nabla_y, \nabla_x] + \nabla_{[x,y]} = 0$, and then $K(\sigma) = 0$.
2. $d\mathbf{x}_p \colon T_p\mathbb{R}^2 \to T_{\mathbf{x}(p)}\mathbb{R}^4$, given by

$$\frac{1}{\sqrt{2}}\begin{pmatrix} -\sin\theta & 0 \\ \cos\theta & 0 \\ 0 & -\sin\varphi \\ 0 & \cos\varphi \end{pmatrix},$$

   is injective, and the composite $\pi \circ d\mathbf{x}_p \colon T_p\mathbb{R}^2 \to T_{\mathbf{x}(p)}\mathbb{R}^4 \to T_{\mathbf{x}(p)}S^3$ (π is the projection) is also injective, since the vector $d\mathbf{x}_p(v)$ is perpendicular to $\overrightarrow{Op} = \frac{1}{\sqrt{2}}(\cos\theta, \sin\theta, \cos\varphi, \sin\varphi)$, and as a result, $d\mathbf{x}_p(v)$ lies in $T_{\mathbf{x}(p)}S^3$ for $v \in T_p\mathbb{R}^2$ and then $\pi = \mathrm{id}$.

   The zero sectional curvature is simply from $\mathbf{x}(\mathbb{R}^2) = S^1\left(\frac{1}{\sqrt{2}}\right) \times S^1\left(\frac{1}{\sqrt{2}}\right)$ and Exercise 1.c).
3. Use the fact that N is totally geodesic in M iff all geodesics in N are also geodesics in M.

4. For $x = x_1 + x_2, y = y_1 + y_2 \in T_{p_1}N_1 \oplus T_{p_2}N_2$, from Exercise 1.a) we have

$$B(x,y) = \overline{\nabla}_{\overline{X}}\overline{Y} - \nabla_X Y = \overline{\nabla}^1_{\overline{X}_1}\overline{Y}_1 + \overline{\nabla}^2_{\overline{X}_2}\overline{Y}_2 - \nabla^1_{X_1}Y_1 - \nabla^2_{X_2}Y_2 = B_1(x_1,y_1) + B_2(x_2,y_2).$$

Then for $\eta = \eta_1 + \eta_2 \in (T_{p_1}N_1)^\perp \oplus (T_{p_2}N_2)^\perp$, we have

$$H_\eta(x,y) = H_{\eta_1}(x_1,y_1) + H_{\eta_2}(x_2,y_2),$$

so H_{η_1}, H_{η_2} vanish implies that H_η vanishes.

5. Let $X_1 + Y_1, X_2 + Y_2 \in \mathscr{X}(S^2) \oplus \mathscr{X}(S^2) = \mathscr{X}(S^2 \times S^2)$, then

$$R(X_i, X_j, X_k, Y_\ell) = R(X_i, X_j, Y_k, Y_\ell) = R(X_i, Y_j, Y_k, Y_\ell) = 0$$

for all $i,j,k,\ell \in \{1,2\}$. The reason is that $g_{X_iY_j} = 0$ for all $i,j \in \{1,2\}$, and thus $\nabla_{X_i}Y_j = 0$. Expand the curvature below, we have

$$R(X_1+Y_1, X_2+Y_2, X_1+Y_1, X_2+Y_2) = R(X_1,X_2,X_1,X_2) + R(Y_1,Y_2,Y_1,Y_2) \geq 0$$

as $K(S^2) = 1 > 0$.

Let i be the inclusion from S^1 into an equator of S^2. By Exercise 4, $(i,i)\colon T^2 = S^1 \times S^1 \hookrightarrow S^2 \times S^2$ is totally geodesic.

7. For any $p \in M$ and $\eta \in (T_pM)^\perp$, $H_\eta(X,Y) = \langle B(X,Y), \eta\rangle \equiv 0$, which is to say, $B(X,Y) = \overline{\nabla}_{\overline{X}}\overline{Y} - \nabla_X Y \equiv 0$.

8. a) Note that $e_1 = \sqrt{2}\frac{\partial \mathbf{x}}{\partial\theta}, e_2 = \sqrt{2}\frac{\partial \mathbf{x}}{\partial\varphi}$, and therefore $e_1, e_2 \in T_pT^2$. And check without effort that e_1, e_2, n_1, n_2 form an orthonormal basis of $T_p\mathbb{R}^4$.

b) Extend $e_1 \in T_pT^2$ to $\sqrt{2}(-x_2\partial_1 + x_1\partial_2) \in T_p\mathbb{R}^4$, and $e_2 \in T_pT^2$ to $\sqrt{2}(-x_4\partial_3 + x_3\partial_4) \in T_p\mathbb{R}^4$. As $\overline{\nabla}_X Y = \sum\limits_k X(y_k)\partial_k$ for $Y = \sum\limits_k y_k\partial_k$ in $\mathbb{R}^4$, we have

$$\begin{aligned}\overline{\nabla}_{e_1}e_1 &= 2\big([(-x_2\partial_1 + x_1\partial_2)(-x_2)]\partial_1 + [(-x_2\partial_1 + x_1\partial_2)(x_1)]\partial_2\big)\\ &= 2(-x_1\partial_1 - x_2\partial_2) = -n_1 + n_2,\end{aligned}$$

and $\overline{\nabla}_{e_1}e_2 = 0$, $\overline{\nabla}_{e_2}e_1 = 0$, $\overline{\nabla}_{e_2}e_2 = 2(-x_3\partial_3 - x_4\partial_4) = -n_1 - n_2$. Then

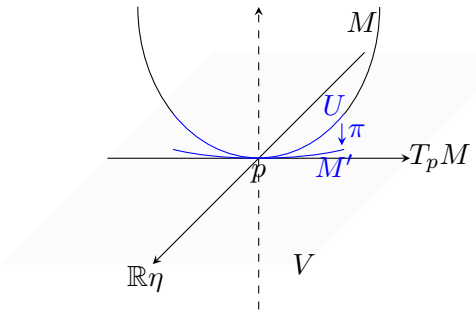
$$\left.\begin{aligned}\langle S_{n_1}e_1, e_1\rangle &= \langle\overline{\nabla}_{e_1}e_1, n_1\rangle = -1,\\ \langle S_{n_1}e_1, e_2\rangle &= \langle\overline{\nabla}_{e_1}e_2, n_1\rangle = 0\end{aligned}\right\} \Rightarrow S_{n_1}e_1 = -e_1;$$

$$\left.\begin{aligned}\langle S_{n_1}e_2, e_1\rangle &= \langle\overline{\nabla}_{e_2}e_1, n_1\rangle = 0,\\ \langle S_{n_1}e_2, e_2\rangle &= \langle\overline{\nabla}_{e_2}e_2, n_1\rangle = -1\end{aligned}\right\} \Rightarrow S_{n_1}e_2 = -e_2;$$

$$\left.\begin{aligned}\langle S_{n_2}e_1, e_1\rangle &= \langle\overline{\nabla}_{e_1}e_1, n_2\rangle = 1,\\ \langle S_{n_2}e_1, e_2\rangle &= \langle\overline{\nabla}_{e_1}e_2, n_2\rangle = 0\end{aligned}\right\} \Rightarrow S_{n_2}e_1 = e_1;$$

$$\left.\begin{aligned}\langle S_{n_2}e_2, e_1\rangle &= \langle\overline{\nabla}_{e_2}e_1, n_2\rangle = 0,\\ \langle S_{n_2}e_2, e_2\rangle &= \langle\overline{\nabla}_{e_2}e_2, n_2\rangle = -1\end{aligned}\right\} \Rightarrow S_{n_2}e_2 = -e_2.$$

c) Since $T_pS^3(1) \oplus \mathbb{R}n_1 = T_p\mathbb{R}^4$ and $T_pT^2 \oplus \mathbb{R}n_2 = T_pS^3(1)$, the mean curvature of $\mathbf{x}$ is $\frac{1}{2}\operatorname{trace} S_{n_2} = \frac{1}{2}(1-1) = 0$.



9. Follow the hint, and a visualization is offered here.
10. Differentiating $\langle S_\eta X, Y\rangle = \langle X, S_\eta Y\rangle$ with respect to V, we obtain

$$\langle \nabla_V(S_\eta X), Y\rangle + \langle S_\eta X, \nabla_V Y\rangle = \langle \nabla_V X, S_\eta Y\rangle + \langle X, \nabla_V(S_\eta Y)\rangle.$$

Using the fact that

$$\begin{aligned}(\nabla_V S_\eta)(X,Y) &= (\nabla S_\eta)(X,Y,V)\\ &= V(S_\eta(X,Y)) - S_\eta(X,\nabla_V Y) - S_\eta(\nabla_V X, Y)\\ &= V\langle S_\eta X, Y\rangle - \langle S_\eta X, \nabla_V Y\rangle - \langle \nabla_V X, S_\eta Y\rangle\\ &= \langle \nabla_V(S_\eta X), Y\rangle - \langle \nabla_V X, S_\eta Y\rangle\end{aligned}$$

and the previous expression, we obtain that

$$(\nabla_V S_\eta)(X,Y) = (\nabla_V S_\eta)(Y,X).$$

11. a) By definition,

$$\begin{aligned}\overline{\triangle} f &= \text{div grad } f\\ &= \text{trace of the linear mapping } Y \mapsto \nabla_Y \text{ grad } f\\ &= \text{trace Hess } f.\end{aligned}$$

b) Use $\overline{\nabla}_X Y - \overline{\nabla}_Y X = XY - YX$ and

$$\begin{aligned}\langle(\text{Hess } f)Y, X\rangle &= \langle \overline{\nabla}_Y \text{ grad } f, X\rangle\\ &= Y\langle \text{grad } f, X\rangle - \langle \text{grad } f, \overline{\nabla}_Y X\rangle\\ &= Y(df(X)) - df(\overline{\nabla}_Y X)\\ &= (YX - \overline{\nabla}_Y X)f.\end{aligned}$$

c) Follow the hint, where $\langle \overline{\nabla}_\eta \eta, \eta\rangle = \eta\langle \eta, \eta\rangle/2 = 0$.

d) We need to verify that grad $f(p) \perp T_pM$. In fact, for any $v \in T_pM$, let γ be a curve in M with $\gamma(0) = p, \gamma'(0) = v$, and then $f \circ \gamma = a$. It follows that $\langle \text{grad } f(p), v\rangle = df_p(v) = \frac{d}{dt}\big|_{t=0}(f\circ\gamma) = 0$.

Complete Manifolds; Hopf–Rinow and Hadamard Theorems

1. Take M to be a non–convex subset of $N = \mathbb{R}^n$.

7. For any curve γ joining $p, q \in M$, $f \circ \gamma$ is a curve joining $f(p), f(q) \in \overline{M}$, then
$$\int |\gamma'| \geq c \int |df(\gamma')| = c \int |(f \circ \gamma)'| \geq cd(f(p), f(q)).$$
Taking infimum on γ, we obtain that
$$d(p, q) \geq cd(f(p), f(q)).$$
We are now equipped to verify that M is complete as a metric space. Let $\{p_n\}$ be a Cauchy sequence in M, that is, $d(p_n, p_m) \to 0$. Then $d(f(p_n), f(p_m)) \to 0$ too. As $\overline{M}$ is complete, $\{f(p_n)\}$ converges to some point in $\overline{M}$, written $f(p)$, for f is surjective. Then $\{p_n\}$ converges to p, as f is a diffeomorphism.
8. Were f not injective, i.e. $f(p) = f(q)$ for two distinct points $p, q \in M$, then for a geodesic γ joining p to q, $f \circ \gamma$ is a loop. Because f is a local isometry, $f \circ \gamma$ is a geodesic too. Take any point r different from $f(p)$ in the image of $f \circ \gamma$, then there are two geodesics joining $f(p)$ to r, a contradiction.
Next we show that f is surjective. For any $s \in \overline{M}$, let $\bar{\gamma}$ be the geodesic segment joining $f(p)$ to s, where $f(p)$ is any point in the image of f. Let γ be the geodesic starting from p with initial speed v such that $df_p(v) = \bar{\gamma}'(0)$. Then $\bar{\gamma}$ is precisely $f \circ \gamma$ as both are geodesics that start from $f(p)$ with the same initial speed, and it follows that $(f \circ \gamma)(l(\bar{\gamma})) = s$, where owing to the completeness of M, γ is defined on the entire $[0, \infty)$, and in particular at $l(\bar{\gamma})$.
Here we used the fact that a local isometry $M \to f(M)$ maps geodesics γ to geodesics $f \circ \gamma$ (mentioned on p.73). The reason: The geodesic equation depends on the metric and its first derivatives in a coordinate chart. A local isometry preserves the metric and is a local diffeomorphism. So it induces coordinates on $f(M)$ with the same metric coefficients. In particular, it must take geodesics to geodesics.
9. The length is
$$\int_\varepsilon^1 \sqrt{\frac{1}{y}} dy = 2\sqrt{y}\Big|_\varepsilon^1 = 2 - 2\sqrt{\varepsilon} \to 2.$$
Note that $x = 0, \varepsilon \leq y \leq 1$ is a geodesic that does not extend beyond 2, and therefore M is not complete.
10. Method I. As illustrated by Chapter 3 Example 3.10 (p.73), all the geodesics in the Lobatchevski plane are of the form $x = x_0, y > 0$ or $x = x_0 + r\cos\theta, y = r\sin\theta, \theta \in (0, \pi)$. Their lengths are respectively
$$\int_0^\infty \frac{dy}{y} = \infty \quad \text{or} \quad \int_\frown \frac{\sqrt{dx^2 + dy^2}}{y} = \int_0^\pi \frac{r d\theta}{r\sin\theta} \geq \int_0^\pi \frac{d\theta}{\min(\theta, \pi - \theta)} = \infty.$$
Method II. Exhaust $\mathbb{R}^2_+$ by a sequence of compact sets $K_n = [-n\ln n, n\ln n] \times [1/n, n]$. Let $p = (0, 1)$, then for $q_n \notin K_n$, it is not hard to show that
$$d(p, q_n) > \min\left(\frac{n\ln n}{n}, \int_{1/n}^1 \frac{dy}{y}, \int_1^n \frac{dy}{y}\right) = \ln n \to \infty.$$

Spaces of Constant Curvature

1. a) Recall that by Chapter 4 Corollary 3.5 (p.96), the metric having constant curvature K is equivalent to $R_{ijij} = -R_{ijji} = K$ for all $i \neq j$ and $R_{ijk\ell} = 0$ in the other cases. From the calculation at the

end of p.161 and

$$f_i = \frac{F_i}{F}, \quad f_{ii} = \frac{F_{ii}F - F_i^2}{F^2},$$

we get

$$R_{ijij} = -\sum_\ell (F_\ell)^2 + (F_{ii} + F_{jj})F;$$

from the formula (2) on the same page,

$$R_{ijk\ell} = R^s_{ijk}\frac{\delta_{s\ell}}{F^2} = \frac{R^\ell_{ijk}}{F^2} = \frac{(-f_k f_j - f_{kj})\delta_{i\ell} + (f_i f_k + f_{ik})\delta_{j\ell}}{F^2} = \frac{-F_{kj}\delta_{i\ell} + F_{ik}\delta_{j\ell}}{F^4},$$

taking $i \neq j = k = \ell$, we get $F_{ij} = 0$ provided that $R_{ijk\ell} = 0$.

b) i. $F_{ij} = 0$ for $i \neq j$ implies that F_i depends only on x_i for all $1 \leq i \leq n$. Then for $x = (x_1, \cdots, x_n)$,

$$F(x) = F(0) + \int_0^{(x_1,0,\cdots,0)} F_1 + \cdots + \int_{(x_1,\cdots,x_{n-1},0)}^{(x_1,\cdots,x_{n-1},x_n)} F_n := G_1(x_1) + \cdots + G_n(x_n).$$

ii. Note that $(F_{ii})_j = (F_{ij})_i = 0$, denote $F_{ii} \equiv 2a$, where each a for $i = 1, \cdots, n$ coincides due to the second equation in a). Then we write

$$F = ax_i^2 + b_i x_i + c_i(x_1, \cdots, \hat{x}_i, \cdots, x_n),$$

where the $\hat{}$ means h_i is a function independent of x_i. Further, express F as

$$F = \sum_i (ax_i^2 + b_i x_i) + c(x_1, \cdots, x_n)$$

and make a comparison (n comparisons indeed), we find that c does not depend on x_i for each $i = 1, \cdots, n$ indeed. Therefore

$$F = \sum_i (ax_i^2 + b_i x_i) + c,$$

and substituting into the second equation in a) yields

$$\Big(\sum_i (ax_i^2 + b_i x_i) + c\Big)4a = K + \sum_{i=1}^n (2ax_i + b_i)^2.$$

c) This is a statement, instead of a problem.

d) For any $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$, the distance between 0 and x is bounded by the length of the curve $\gamma = \{tx : 0 \leq t \leq 1\}$, which is

$$\begin{aligned} d(0,x) &\leq \int_\gamma |\gamma'|_g = \int_\gamma \frac{1}{1 + \frac{K}{4}|tx|^2}|x| \\ &= \int_0^1 \frac{d(t|x|)}{1 + \left(\frac{\sqrt{K}}{2}t|x|\right)^2} \\ &= \frac{2}{\sqrt{K}} \arctan\left(\frac{\sqrt{K}}{2}|x|\right) < \frac{\pi}{\sqrt{K}}. \end{aligned}$$

Hence $(\mathbb{R}^n, g)$ is bounded, and also closed as it is the whole space. But $\mathbb{R}^n$ is non–compact (compactness is a topological property), violating b) of Hopf–Rinow Theorem. Then $(\mathbb{R}^n, g)$ is not complete.

13. It is clear that $\sigma(\gamma(t)) = \gamma(-t)$, as both are geodesics with the same starting point p and initial speed $-\gamma'(t)$. To prove that $d\sigma_{\gamma(t)}X(\gamma(t))$ is parallel along $\gamma(t)$, compare

$$\frac{d}{dt}\langle X, Y\rangle_{\gamma(t)} = \langle X, \nabla_{\dot\gamma} Y\rangle_{\gamma(t)} + \langle \nabla_{\dot\gamma} X, Y\rangle_{\gamma(t)} = \langle X, \nabla_{\dot\gamma} Y\rangle_{\gamma(t)}$$

and

$$\begin{aligned}\frac{d}{dt}\langle d\sigma(X), d\sigma(Y)\rangle_{\sigma(\gamma(t))} &= \langle d\sigma(X), \nabla_{d\sigma(\dot\gamma)} d\sigma(Y)\rangle_{\sigma(\gamma(t))} + \langle \nabla_{d\sigma(\dot\gamma)} X, d\sigma(Y)\rangle_{\sigma(\gamma(t))} \\ &= \langle X, \nabla_{\dot\gamma} Y\rangle_{\gamma(t)} + \langle \nabla_{d\sigma(\dot\gamma)} X, d\sigma(Y)\rangle_{\sigma(\gamma(t))},\end{aligned}$$

we get the last term is 0, and consequently $\nabla_{d\sigma(\dot\gamma)} X(\sigma(\gamma(t))) = 0$ by the arbitrariness of Y (and that $d\sigma$ is surjective). Note that at the colored "=", $d\sigma(\nabla_{\dot\gamma} Y) = \nabla_{d\sigma(\dot\gamma)} d\sigma(Y)$ is used (check it using (9) of Chapter 2 Theorem 3.6, p.55).

Variations of Energy

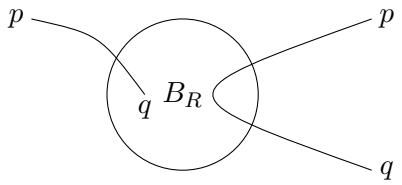
1. Suppose that $\{q_n\}$ is a sequence of points in N such that $d(p_0, q_n) \to d(p_0, N)$, then it follows from the boundedness and thus compactness of $\{q_n\}$ that there is a cluster point q_0, and $q_0 \in N$ as N is closed. Let $\gamma\colon [0, \ell] \to M$ be a minimizing geodesic joining p_0 to q_0. Consider the variational fields $V(t)$ of variations keeping the starting point p_0 fixed yet with varying end points in N. As $\ddot\gamma = 0$ and $V(0) = 0$, we have

$$0 = \frac{1}{2}E'(0) = -\int_0^\ell \langle V, \ddot\gamma\rangle - \langle V(0), \dot\gamma(0)\rangle + \langle V(\ell), \dot\gamma(\ell)\rangle = \langle V(\ell), \dot\gamma(\ell)\rangle,$$

then $\dot\gamma(\ell)$ is perpendicular to all $V(\ell) \in T_{q_0}N$.

2. Suppose, to the contrary, that $K(x, y) \geq \frac{1}{\kappa^2} > 0$ on $\{(x, y) : x^2 + y^2 \geq R^2\}$ for a certain $R > 0$. As will be soon shown, $\operatorname{diam} \mathbb{R}^2 \leq 2\pi\kappa + \operatorname{diam} B_R$, where $B_R = \{(x, y) : x^2 + y^2 \leq R^2\}$ is compact hence bounded. Then the whole $\mathbb{R}^2$ is bounded and hence compact, a contradiction.

   For any $p, q \in \mathbb{R}^2$, join them by a minimizing geodesic γ and establish a uniform upper bound of its length. If $\gamma \subset B_R$, then $\ell(\gamma) \leq \operatorname{diam} B_R$; if $\gamma \subset B_R^c$, then $\ell(\gamma) \leq \pi\kappa$ as the proof of Bonnet–Myers theorem (Theorem 3.1) indicates. Things are worse if $p \notin B_R, q \in B_R$, in which case $\ell(\gamma) \leq \pi\kappa + \operatorname{diam} B_R$; the worst case is when γ intersects $\{(x, y) : x^2 + y^2 = R^2\}$ twice (had it crossed the boundary more than twice, it would not be the shortest), and $\ell(\gamma) \leq 2\pi\kappa + \operatorname{diam} B_R$.



3. Suppose that the geodesic is of length ℓ, then

$$\begin{aligned}0 \leq \sum_{j=1}^{n-1} E_j''(0) &\leq \int_0^\ell \sin^2\frac{\pi s}{\ell}\Big[(n-1)\Big(\frac{\pi}{\ell}\Big)^2 - (n-1)\Big(a + \frac{df}{ds}\Big)\Big]ds \\ &= (n-1)\int_0^\ell \sin^2\frac{\pi s}{\ell}\Big[\Big(\frac{\pi}{\ell}\Big)^2 - a - \frac{df}{ds}\Big]ds\end{aligned}$$

$$= (n-1)\left[\frac{\ell}{2}\Big[\Big(\frac{\pi}{\ell}\Big)^2 - a\Big] + \int_0^\ell \frac{\pi}{\ell}\sin\frac{2\pi s}{\ell} f\, ds\right]$$
$$\le (n-1)\left(\frac{\pi^2}{2\ell} - \frac{a\ell}{2} - 2c\right).$$

So

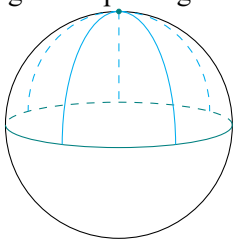
$$\ell \le \frac{2c + \sqrt{4c^2 + a\pi^2}}{a}.$$

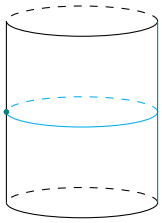
4. Suppose that $P_{\gamma(t)}$ is the parallel transport along γ from $\gamma(0)$ to $\gamma(t)$, and denote $P_{\gamma(2\pi)}$ by P, where γ is assumed to be parametrized by $[0, 2\pi)$ (so $\gamma(0) = \gamma(2\pi)$). P preserves the inner product and is thus orthogonal; M is orientable and n is even, so $\det P = 1 = (-1)^n$. Applying Lemma 3.8 (p.203), P leaves invariant some nonzero $v \perp \dot\gamma(0)$, that is, $Pv = v \in T_{\gamma(2\pi)}M = T_{\gamma(0)}M$. Then $V(t) := P_{\gamma(t)}$ is a vector field along the closed curve γ, and $Pv = v$ ensures that it is well–defined. Now
$$\frac{1}{2}E_V''(0) = -\int_0^{2\pi} \langle V, R(\dot\gamma, V)\dot\gamma\rangle < 0.$$
Note that though a variation that generates V is improper, the last four terms of (5) of this chapter (p.199) all offset as γ is a closed curve.

5. a) Existence is from compactness, and using Exercise 1 twice we get $\gamma \perp N_1, \gamma \perp N_2$.
b) Denote $\nabla, \nabla^{(1)}, \nabla^{(2)}$ the Riemannian connections on M, N_1, N_2, and write h' for $\frac{\partial h}{\partial s}$. We have
$$\begin{aligned}\frac{1}{2}E''(0) &= I_\ell(V,V) + \left\langle \frac{D}{ds}\frac{\partial h}{\partial s}, \dot\gamma\right\rangle(0,\ell) - \left\langle \frac{D}{ds}\frac{\partial h}{\partial s}, \dot\gamma\right\rangle(0,0)\\ &= I_\ell(V,V) + \langle \nabla_{h'}h', \dot\gamma\rangle(0,\ell) - \langle \nabla_{h'}h', \dot\gamma\rangle(0,0)\\ &= I_\ell(V,V) + \langle B(h',h'), \dot\gamma\rangle(0,\ell) - \langle B(h',h'), \dot\gamma\rangle(0,0)\\ &= I_\ell(V,V) + \langle V(\ell), S^{(2)}_{\dot\gamma(\ell)}V(\ell)\rangle - \langle V(0), S^{(1)}_{\dot\gamma(0)}V(0)\rangle.\end{aligned}$$
Here we used $h'(\ell) \in T_{p_2}N_2, h'(0) \in T_{p_1}N_1$ where p_1, p_2 are the two endpoints of γ, and hence $\nabla^{(2)}_{h'(\ell)}h'(\ell) \in T_{p_2}N_2 \perp \dot\gamma(\ell), \nabla^{(1)}_{h'(0)}h'(0) \in T_{p_1}N_1 \perp \dot\gamma(0)$.

6. a) (i) Since $\ddot\gamma \equiv 0, V(0) = 0$,
$$\frac{1}{2}E'(s) = -\int_0^\ell \langle V, \ddot\gamma\rangle - \langle V(0), \dot\gamma(0)\rangle + \langle V(s), \dot\gamma(s)\rangle = \langle V(s), \dot\gamma(s)\rangle = \langle \sigma_s'(d(s)), \dot\gamma(s)\rangle.$$
(ii) Put $V(t) = \frac{\partial h}{\partial s}(s_0, t)$. Then
$$\frac{1}{2}E''(s_0) = I_{d(s_0)}(V,V) - \left\langle \frac{D}{ds}\frac{\partial h}{\partial s}, \dot\gamma\right\rangle(s_0,0) + \left\langle \frac{D}{ds}\frac{\partial h}{\partial s}, \dot\gamma\right\rangle(s_0, d(s_0)) = I_{d(s_0)}(V,V) \ge 0,$$
where we used $\frac{\partial h}{\partial s}(s_0,0) = 0$ as $h(s_0,0) = \sigma_{s_0}(0) \equiv p$, and $\frac{D}{ds}\frac{\partial h}{\partial s}(s_0, d(s_0)) = \frac{D}{ds}\frac{\partial \gamma}{\partial s}(s_0) = 0$ as γ is a geodesic. Suppose that $E''(s_0) = 0$, then combined with $K \le 0$ we get $\dot V(t) \equiv 0$, so $V(t) \equiv 0$ as a result of $V(0) = 0$.
b) By Hadamard's theorem (3.1 of Chapter 7), $\exp_p\colon T_p\tilde M \to \tilde M$ is a diffeomorphism. Hence every geodesic joining p to any point $\gamma(s)$ of γ is a curve in the variation h (where s is assumed to range over the entire $(-\infty, \infty)$). Use the property of convex functions.
c) When $K \le 0$ is dropped, let $\tilde M$ be a sphere, p be the north pole and γ be its equator, then there are infinitely many perpendicular lines of minimal length.

When simple connectedness is dropped, let $\tilde{M}$ be an infinite cylinder and p be any point, and γ be the straight line passing through the antipodal point of p, then there are two lines reaching minimal length.





The Rauch Comparison Theorem

2. Take $K_0 > 0$ arbitrarily small in Klingenberg's lemma and show that if M is simply connected, there exists a unique geodesic joining the two points $p, q \in M$. This is because were there to exist two such geodesics α_0 and α_1, they would be homotopic by the simple connectedness, and by Klingenberg's lemma, there would exist a sequence of curves α_t in this homotopy of lengths $\geq \pi/\sqrt{K_0}$, which would be arbitrarily big; however, $\ell(\alpha_t)$ as a continuous function of $t \in [0,1]$ is bounded, a contradiction.

   This asserts that $\exp_p \colon T_pM \to M$ is a one-one map. That is, it has an inverse $\exp_p^{-1}$. It suffices to show that $\exp_p$ is a local diffeomorphism then, as this will imply that $\exp_p^{-1}$ is continuous locally, and hence globally. This is a consequence of Rauch's Theorem, in combination with Chapter 5 Proposition 3.5 (p.117) (and Chapter 0 Theorem 2.10, p.10).
3. Note that $\tilde{\gamma}(t) = tv$ is a geodesic on T_pM, and $\tilde{J}(t) = tw$ is a Jacobi field along $\tilde{\gamma}$; further, $J(t) = (d\exp_p)_{tv}(tw)$ is a Jacobi field along γ. As

$$J(0) = \tilde{J}(0) = 0, \quad J'(0) = \tilde{J}'(0) = w, \quad \gamma'(0) = \tilde{\gamma}'(0) = v,$$

   and $K \leq 0 = \tilde{K}$, we have $|J(1)| \geq |\tilde{J}(1)|$, which is $|(d\exp_p)_v(w)| \geq |w|$.
5. a) For all $t \in [0, \ell]$, integration by parts gives

$$\begin{aligned} 0 &= \int_0^t \tilde{f}(f'' + Kf) - f(\tilde{f}'' + \tilde{K}\tilde{f}) \\ &= [\tilde{f}f' - f\tilde{f}']_0^t + \int_0^t (K - \tilde{K}) f\tilde{f}. \end{aligned}$$

   From the initial conditions $f(0) = 0$ and $f'(0) = 1$, f is positive in a neighborhood of 0. Assume that $f(t_1) = 0$ for certain $t_1 < t_0$. Then take $t = t_1$ and get

$$0 = \big(\underbrace{\tilde{f}(t_1)}_{>0}\underbrace{f'(t_1)}_{\leq 0} - \underbrace{f(t_1)}_{=0}\tilde{f}'(t_1)\big) - \big(\underbrace{\tilde{f}(0)}_{=0}f'(0) - \underbrace{f(0)}_{=0}\tilde{f}'(0)\big) + \int_0^{t_1} (\underbrace{K - \tilde{K}}_{\leq 0})\underbrace{f\tilde{f}}_{>0},$$

   forcing $K \equiv \tilde{K}$. Then f and $\tilde{f}$ satisfy exactly the same ode, with the same initial conditions $f(0) = \tilde{f}(0) = 0$ and $f'(0) = \tilde{f}'(0) = 1$, and for this reason $f \equiv \tilde{f}$.

   b) Again, examine

$$0 = \big(\tilde{f}(t)f'(t) - f(t)\tilde{f}'(t)\big) - \big(\underbrace{\tilde{f}(0)}_{=0}f'(0) - \underbrace{f(0)}_{=0}\tilde{f}'(0)\big) + \int_0^t (\underbrace{K - \tilde{K}}_{\leq 0})\underbrace{f\tilde{f}}_{>0}.$$

It follows that $\tilde{f}f' - f\tilde{f}' \geq 0$, that is, $(f/\tilde{f})' = (\tilde{f}f' - f\tilde{f}')/(\tilde{f}^2) \geq 0$. Conclude by

$$f/\tilde{f} \geq \lim_{t\to 0}(f/\tilde{f}) = \lim_{t\to 0}(f'/\tilde{f}') = 1.$$

For equality, $\tilde{f}f' - f\tilde{f}' \equiv 0$ forces $K \equiv \tilde{K}$ as a).

To verify the Theorem of Rauch in dimension two as an application, first reduce the Jacobi fields in question to normal Jacobi fields as the proof in the book does, and hence suppose that $J(t) = f(t)e_2(t)$ where $e_2(t)$ is the parallel transport of a unit vector $e_2 \in T_{\gamma(0)}M$ with $e_2 \perp \gamma'(0)$. Then $\frac{D^2J}{dt^2} = f''e_2$, so the Jacobi equation is equivalent to

$$\langle f''e_2 + R(\gamma', fe_2)\gamma', \gamma'\rangle = 0$$

and

$$\langle f''e_2 + R(\gamma', fe_2)\gamma', e_2\rangle = 0,$$

since $\{\gamma', e_2\}$ spans $T_{\gamma(t)}M$. The first equation trivially holds, and the second is $f'' + Kf = 0$. Similarly for $\tilde{J}(t)$.

6. Suppose that $\tilde{f}(t) \neq 0$ for all $t \in [0, t_0]$. Using the expression from the last exercise,

$$\int_0^{t_0} \underbrace{(K-L)}_{\leq 0}\underbrace{f\tilde{f}}_{<0} + \underbrace{\tilde{f}(t_0)}_{>0}\underbrace{f'(t_0)}_{\geq 0} - \underbrace{\tilde{f}(0)}_{>0}\underbrace{f'(0)}_{\leq 0} = 0.$$

This forces $K \equiv L$, as well as $f'(0) = f'(t_0) = 0$. Then the initial conditions $f(0) = f'(0) = 0$ imply that $f \equiv 0$, contradicting that $J \neq 0$ on $(0, t_0)$.

7. A correction: it should be

$$\left|\int_t^\infty K(s)ds\right| \leq \frac{1}{4(t+1)}.$$

a)

$$\omega^2 \leq \left(\frac{1}{4(t+1)} + \frac{1}{4(t+1)}\right)^2 = \left(-\frac{1}{4(t+1)}\right)' = -\omega' - K.$$

b) $\tilde{f}' = \omega\tilde{f}$, and $\tilde{f}'' = \omega'\tilde{f} + \omega\tilde{f}' = (\omega' + \omega^2)\tilde{f} = -L\tilde{f}$. $\tilde{f}(0) = \exp 0 = 1$.

c) $-L = \omega' + \omega^2 \leq -K$, which is $K \leq L$. Then the oscillation theorem of Sturm yields that $\tilde{f}$ has a zero on $[0, s_0]$. But $\tilde{f} > 0$.

The Morse Index Theorem

2. Expand f and f' into the Fourier series

$$f(t) = \sum_{n=1}^\infty a_n \sin nt, \quad f'(t) = \sum_{n=1}^\infty na_n \cos nt.$$

Then the inequality follows from

$$\int_0^\pi f^2 = \frac{\pi}{2}\sum_{n=1}^\infty a_n^2, \quad \int_0^\pi f'^2 = \frac{\pi}{2}\sum_{n=1}^\infty n^2a_n^2.$$

Yet we need to justify that differentiating f by terms does produce f', which relies on the decay of the coefficients a_n. Indeed, integrating by parts, we find

$$\frac{\pi}{2}a_n = \int_0^\pi f\sin nt = \int_0^\pi f'\frac{\cos nt}{n} = -\int_0^\pi f''\frac{\sin nt}{n^2}.$$

Then

$$\left(\sum_{n=1}^{\infty} na_n\right)^2 \le \sum_{n=1}^{\infty}(n^2a_n)^2 \sum_{n=1}^{\infty}\frac{1}{n^2}$$
$$= \left(\frac{\pi}{2}\right)^{-2}\sum_{n=1}^{\infty}\left(\int_0^{\pi} f''\sin nt\right)^2\frac{\pi^2}{6} = \frac{2}{3}\int_0^{\pi} f''^2 < \infty,$$

yielding the uniform convergence of the series of f', and consequently the justification.

3. It follows that $K(t) \equiv 0$ for $t \in (0,\pi)$. Let $t \to 0$, we get $K(p) = 1$. Then use Theorem 4.1 of Chapter 8 (p.163).
4. $\varphi'(0) = 0, \varphi''(0) = -a(0) < 0$ implies that $\varphi'(t_0) < 0$ for some $t_0 > 0$. Suppose that φ has no positive zero, then $\varphi'' = -a\varphi \le 0$ on $(0,+\infty)$. Hence $\varphi'(t) \le \varphi'(t_0)$ for $t > t_0$. But then $\varphi(t) \le \varphi(t_0) + (t-t_0)\varphi'(t_0) \to -\infty$ as $t \to +\infty$, a contradiction.
5. Since $\varphi(-t_1) = \varphi(t_2) = 0$, and $\varphi_Y \ge K \ge a$ with strict inequality at 0,

$$I_{[-t_1,t_2]}(X,X) = \int_{-t_1}^{t_2} \varphi'^2 - \varphi^2\varphi_Y$$
$$= (\varphi\varphi')\big|_{-t_1}^{t_2} - \int_{-t_1}^{t_2}\varphi\varphi'' + \varphi^2\varphi_Y < -\int_{-t_1}^{t_2}\varphi\varphi'' + \varphi^2 a = 0.$$

6. From Exercise 5, there exists $t_0 > 0$ and a vector field Y along the segment $\gamma([-t_0,t_0])$ such that $I(Y,Y) < 0$. Let f be a proper variation that generates Y, then $\frac{1}{2}E''(0) = I(Y,Y) < 0$, which is, $\gamma([-t_0,t_0])$ does not minimize the length between its endpoints.
   If $K \ge 0$ instead, $M = \mathbb{R}\times S^{n-1}$ violates the theorem.
   Remark: Cf. [YS] 1.2 (pp.13–7) for the proof of the splitting theorem (Cheeger–Gromoll, 1971) below.
   Let M be an n-dimensional complete Riemannian manifold with $\mathrm{Ric}(M) \ge 0$. Suppose that M has a line. Then M is isometric to $\mathbb{R}\times N$, where N is an $(n-1)$-dimensional Riemannian manifold with the product metric.

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1.0 试卷: Riemann 几何

原网页

1.1 25 年秋

1. 同 24 秋 1.
2. 同 24 秋 2.
3. 叙述 Hessian 比较定理和 Laplace 比较定理.
4. 计算标准球面和双曲空间距离函数的拉普拉斯.
5. 设 M 是完备 Riemann 流形, Ricci 曲率以 $(n-1)$ 为下界. 若 M 直径等于 π, 证明 M 和单位球面等距.
6. 设 M 是偶数维定向正截面曲率 Riemann 流形, γ 是 M 中闭测地线. 证明: γ 同伦于一条长度严格小于 γ 的闭曲线.
7. 同 24 秋 8.
8. 设 M 是 $\mathbb{R}^{n+1}$ 中极小曲面, 且 M 上存在光滑正函数 f, 满足 $|A|^2 f < -\Delta_M f$, 其中 $|A|^2$ 是第二基本形式, Δ_M 是 Laplace 算子. 证明 M 是稳定的, 即对 M 上任何紧支撑光滑函数 ϕ, 都成立
$$\int_M |A|^2\phi^2 \le \int_M |\nabla_M \phi|^2.$$

1.2 24 年秋

前四题各 12 分，后四题各 13 分。

1. 叙述流形上曲率张量的定义及其基本性质.
2. 叙述割点、共轭点的定义，并写出它们之间的关系.
3. 叙述距离函数的 Laplace 比较定理和 Bishop–Gromov 体积比较定理.
4. 计算标准球面上距离函数的 Hessian 和 Laplace.

5. 根据标准正交基下流形上光滑函数的梯度和 Laplace 的定义，计算曲线坐标系下光滑函数的梯度和 Laplace 的表达式.
6. 设 Riemann 流形 M 的截面曲率 K 满足 $0 < L \leq K \leq H$，其中 H 和 L 是常数。设 γ 是 M 中的测地线，证明：γ 上相邻共轭点沿着 γ 的距离 d 满足

$$\frac{\pi}{\sqrt{H}} \leq d \leq \frac{\pi}{\sqrt{L}}.$$

7. 计算欧氏空间中的极小超曲面上位置函数的 Laplace，并证明欧氏空间中不存在紧（无边）极小超曲面.
8. 证明：具有正截面曲率的流形上不存在 line，即极小测地线 $\gamma\colon (-\infty, +\infty) \to M$. 并判断：对于一个具有非负截面曲率的流形，如果其上有一点的截面曲率为正，是否存在 line.

1.3 24 年春

前四题各 10 分，后四题各 15 分。

1. 叙述 Levi-Civita 联络的定义.
2. 叙述割点、共轭点的定义，并写出它们之间的关系.
3. 计算常曲率空间中距离函数的 Laplacian.
4. 叙述距离函数的 Laplacian 比较定理和 Bishop–Gromov 体积比较定理.
5. 设 M 是非正曲率完备 Riemann 流形, $x \in M$, 证明: $\exp_x\colon \mathrm{T}_xM \to M$ 不减距离, 即对任意 $X \in \mathrm{T}_xM$, $v \in \mathrm{T}_X(\mathrm{T}_xM) \cong \mathrm{T}_xM$, 有 $|(\mathrm{d}\exp_x)_X(v)| \geq |v|$.
6. 设 M 是偶数维正曲率紧 Riemann 流形, 证明: 对 M 上的任意 Killing 向量场 X, 存在 $p \in M$, 使得 $X(p) = 0$.
7. 证明: Clifford 环面 $S^1\left(\frac{1}{\sqrt{2}}\right) \times S^1\left(\frac{1}{\sqrt{2}}\right)$ 是 S^3 中的极小曲面.
8. 设 k 是正整数, 证明：Clifford 环面 $S^k\left(\frac{1}{\sqrt{2}}\right) \times S^k\left(\frac{1}{\sqrt{2}}\right)$ 是 S^{2k+1} 中的极小超曲面. $S^k\left(\frac{1}{\sqrt{2}}\right) \times S^k\left(\frac{1}{\sqrt{2}}\right)$ 是否具有非负截面曲率？请证明.